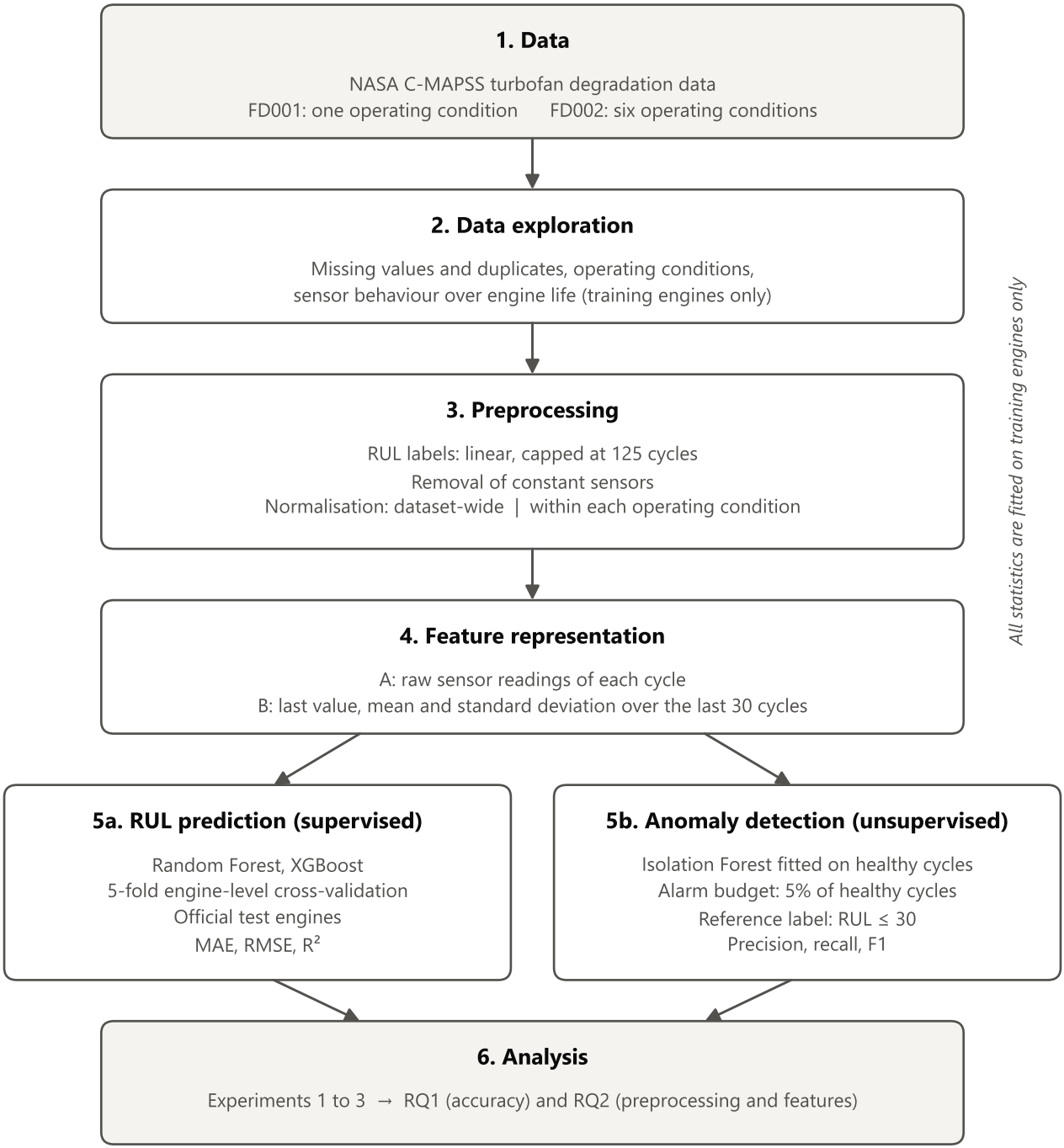
1. Data
NASA C-MAPSS turbofan degradation data
FD001: one operating condition FD002: six operating conditions
2. Data exploration
Missing values and duplicates, operating conditions,
sensor behaviour over engine life (training engines only)
3. Preprocessing
RUL labels: linear, capped at 125 cycles
Removal of constant sensors
Normalisation: dataset-wide | within each operating condition
4. Feature representation
A: raw sensor readings of each cycle
B: last value, mean and standard deviation over the last 30 cycles
5a. RUL prediction (supervised)
Random Forest, XGBoost
5-fold engine-level cross-validation
Official test engines
MAE, RMSE, R²
5b. Anomaly detection (unsupervised)
Isolation Forest fitted on healthy cycles
Alarm budget: 5% of healthy cycles
Reference label: RUL ≤ 30
Precision, recall, F1
6. Analysis
Experiments 1 to 3 → RQ1 (accuracy) and RQ2 (preprocessing and features)
All statistics are fitted on training engines only